How to Create a Tailored Training Using Modular Training Material

This step-by-step guide helps you design and deliver a customized training using the modular training materials.

Step 1: Define the Training Format

Start by clarifying the structural framework of your training.

1.1 Decide on the Delivery Mode

- Will the training be online, in-person, or hybrid?
- Are there technical requirements (platforms, room setup, recording)?

1.2 Define Duration and Structure

- How long will the training last?
- Is it a single session or spread across multiple sessions?

1.3 Clarify the Level of Interaction

- How interactive should the training be? → Will participants engage in discussions, polls, breakout groups, or hands-on exercises?

→ Check for Predefined Requirements

Determine whether the training:

- Is embedded in an existing format (e.g., a colloquium, seasonal school, conference, graduate program)
- Has predefined constraints regarding time, structure, or assessment
- Needs to align with institutional or program-specific guidelines

! Always adapt the format to both the context and the audience !

Step 2: Analyze Your Target Audience

Understanding your participants is essential for tailoring the training.

2.1 Identify Participant Profiles

- Are they data producers, data re-users, or both?
- Are they beginners in research data management (RDM) or more experienced?

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- What types of data categories do they work with (e.g., field data, sensor data, socio-economic data, modeling data)?
- Do they share similar institutional infrastructures?

2.2 Identify Their Needs

- What challenges do they face in agrosystem-specific RDM?
- Do they need practical tool guidance or conceptual understanding?
- Are there regulatory or funder requirements they must meet?

2.3 Consider the Trainers' Expertise

- Who are the trainers?
- What domain expertise and RDM knowledge do they bring?
- Will specific tools or infrastructures be presented?

Step 3: Define the Content Structure

Now, create a rough structure of the content you want to include.

3.1 Select Core Topics and Subtopics

Choose topics that match:

- Audience needs
- Training format
- Available time
- Trainers' expertise

Choose from included topics and subtopics:

01_Researchdatamangement
    01-01_Researchdata
    01-02_Researchdatamanagement
    01-03_DataLifecycle
02_GoodResearchPractice
03_ResearchPolicies
04_FAIR-Principles
05_DataOrganisation
    05-02_DataOrganisation_NamingConvention
    05-03_DataOrganisation_FolderStructure
    05-04_DataOrganisation_VersionControl
06_DataDocumentation
    06-01_DataDocumentation_Intro

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06-02_DataDocumentation_Types
06-03_DataDocumentation_Metadata
06-04_DataDocumentation_MetadataStandards
06-05_DataDocumentation_SemanticWeb
06-06_DataDocumentation_PersistentIdentifier
06-07_DataDocumentation_ComputationalReproducibility
07_DataManagementPlan
07-07_DataManagementPlan_Tools
08_DataCollection
08-01_DataCollection_ElectronicLabNotebooks
08-02_DataCollection_ElectronicFieldbooks
08-03_DataCollection_DataReuse
09_DataStorage
10_DataPublication
10-01_DataPublication_Sharing
10-02_DataPublication_Repositories
10-03_DataPublication_Data-Journals
11_LegalAspects
11-00_LegalAspects_CaseStudy
11-01_LegalAspects_DataRights
11-02_LegalAspects_DataProtection
11-03_LegalAspects_BusinessSecrets
11-04_LegalAspects_Nagoya

3.2 Define the Focus Areas

Not all topics can be covered in depth. Decide:

- Which topics are essential?
- Which can be optional or supplementary?

3.3 Structure the Sequence

Organize topics in a logical order. You may:

- Follow the Research Data Lifecycle
- Use a case study as a guiding example
- Use the included research journey (00-06)

Important considerations:

- Place essential or complex topics earlier in the training.
- Avoid scheduling key content at the very end, when attention may decline.
- Allow buffer time for discussion and questions.

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Step 4: Select Concrete Content and Exercises (Create a Teachingscript)

Once the structure is defined, select the relevant modules (“bricks”) from the modular material.

4.1 Selection of learning objectives for every topic to include

- Select learning objectives you want to include from the relevant topic-teachingscript
- Copy the respective line with all the information of the fitting brick from the topic-teachingscript to your training event-teachingscript
    - you can use the Template_Teachingscript to create this

4.2 Balance Input and Interaction

Alternate between:

- Input phases (“breathing in”): short presentations or explanations
- Reflection and interaction (“breathing out”): discussions, exercises, polls

This improves learning retention and engagement.

4.3 Adapt to Experience Levels

If participants have diverse backgrounds:

- Include exchange formats (peer discussions, experience sharing)

If the group is homogeneous:

- Tailor exercises directly to their specific workflows or data types

If participants are from same institution/working group:

- include material on institutional infrastructures

4.4 Create Framework of your Training Event

- Include welcome-section
    - Create title, introduction of the trainers,iIntroduction of the institution, agenda of the training event
    - include an icebreaker in the beginning
        - this is especially important for interactive courses
- Include question sessions
- Include breaks

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    - make sure to have a short coffee break approximately every ½ to 1 hour
    - make sure to include bigger (lunch) breaks in longer training formats
- Include wrap up
    - Include a wrap-up / take home message section
    - decide if you want to include a section for gaining feedback
    - include further resources and/or further support infrastructure information

4.5 Personalize it for your target audience

- include bricks on institutional infrastructure or domain specific tools or examples that are not included in this material

→ You can now see if the planned timing (Column G in the Teachingscript) fits to your predefined format and framework.

Step 5: Develop the Training Materials

5.1 Compile the Presentation

- Assemble slides from the selected modular "bricks"

5.2 Prepare Interactive Elements

- Create polls (e.g., using online tools)
- Set up breakout group instructions
- Provide templates or worksheets if needed

5.3 Prepare Supporting Materials

- Handouts
- Links to tools and infrastructures
- Case study descriptions
- Further reading resources

→ for online training it may be useful to prepare a document where you can prepare all links to share via chat during the training

→ for in-person training decide if you need participants to bring their laptops (make sure you have power sockets available) - if so it can be helpful to prepare QR Codes AND Short URLs or one shared doc, where all needed resources and links can be found, and that can be shared with the participants in advance

Step 6: Test and Finalize

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Before delivering the training:

- Review timing realistically (adapt our timing suggestions in the teaching script according to your need)
- Test technical tools (especially for online formats)
- Clarify roles if multiple trainers are involved (add responsibilities into column L in the teaching script)
- Prepare a backup plan (e.g., in case of technical issues)

Step 7: Deliver and Reflect

During the Training

- Monitor time using the teaching script
- Be flexible if discussions require more space
- Adjust depth depending on participant feedback

After the Training

- Collect feedback from participants
- Collect feedback from all the trainers and organisers involved
    - Reflect on:
        - What worked well?
        - Which modules were too detailed or too superficial?
        - Were interaction and input well balanced?

Use this reflection to continuously improve future training sessions.

➔ We would also love to receive your feedback!
➔ We would love to hear from you about what you used our material for and how you found it!
➔ Please contact us via: training@fairagro.net

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